

JavaScript Promise Methods — Complete Study List

A. Static Promise Methods

1. `Promise.resolve(value)`

- Creates a fulfilled promise
- Flattens nested promises

2. `Promise.reject(reason)`

- Creates a rejected promise

3. `Promise.all(iterable)`

- Resolves when all promises resolve
- Rejects on first failure

4. `Promise.allSettled(iterable)`

- Waits for all promises
- Never rejects, returns status report

5. `Promise.race(iterable)`

- First settled promise wins

6. `Promise.any(iterable)`

- First fulfilled promise wins
- Rejects with `AggregateError` if all fail

B. Instance Promise Methods

7. `promise.then(onFulfilled, onRejected)`

- Always returns a new promise
- Supports chaining

8. `promise.catch(onRejected)`

- Handles errors from previous steps

9. `promise.finally(onFinally)`

- Executes cleanup logic
- Return value is ignored

C. Async Helpers (Promise-related)

10. async / await

- Syntax sugar over promises

11. queueMicrotask(fn)

- Manually schedule microtasks

D. Core Promise Concepts

12. Promise states

- pending

- fulfilled

- rejected

13. Promise chaining and flattening

14. Error propagation rules

15. Unhandled promise rejections

16. Microtask queue behavior

Suggested Study Order:

1. Promise.resolve

2. Promise.reject

3. then()

4. catch()

5. finally()

6. Promise.all

7. Promise.allSettled

8. Promise.race

9. Promise.any

10. Real-world async patterns